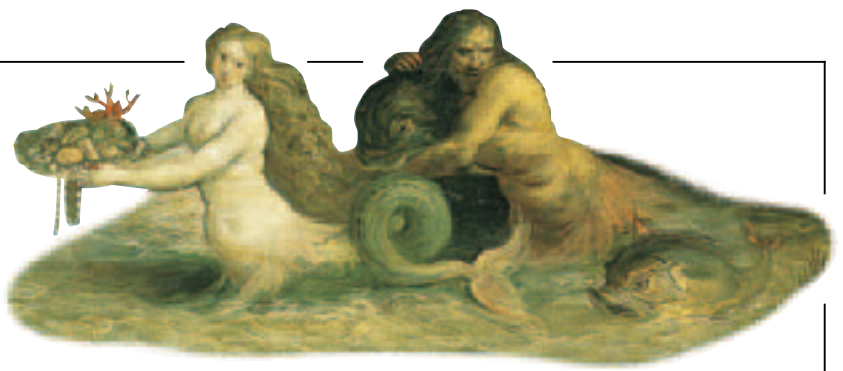




Floating on the Dead Sea

SEA OR LAKE?

The water in the Dead Sea is saltier than any ocean because the water that drains into it evaporates in the hot sun, leaving behind the salts. A body is more buoyant in such salty water, making it easier to float. The Dead Sea is a lake, not a sea, because it is completely surrounded by land. True seas are always connected to the ocean by a channel.

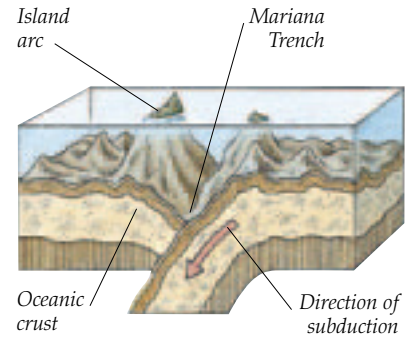


GOD OF THE WATERS

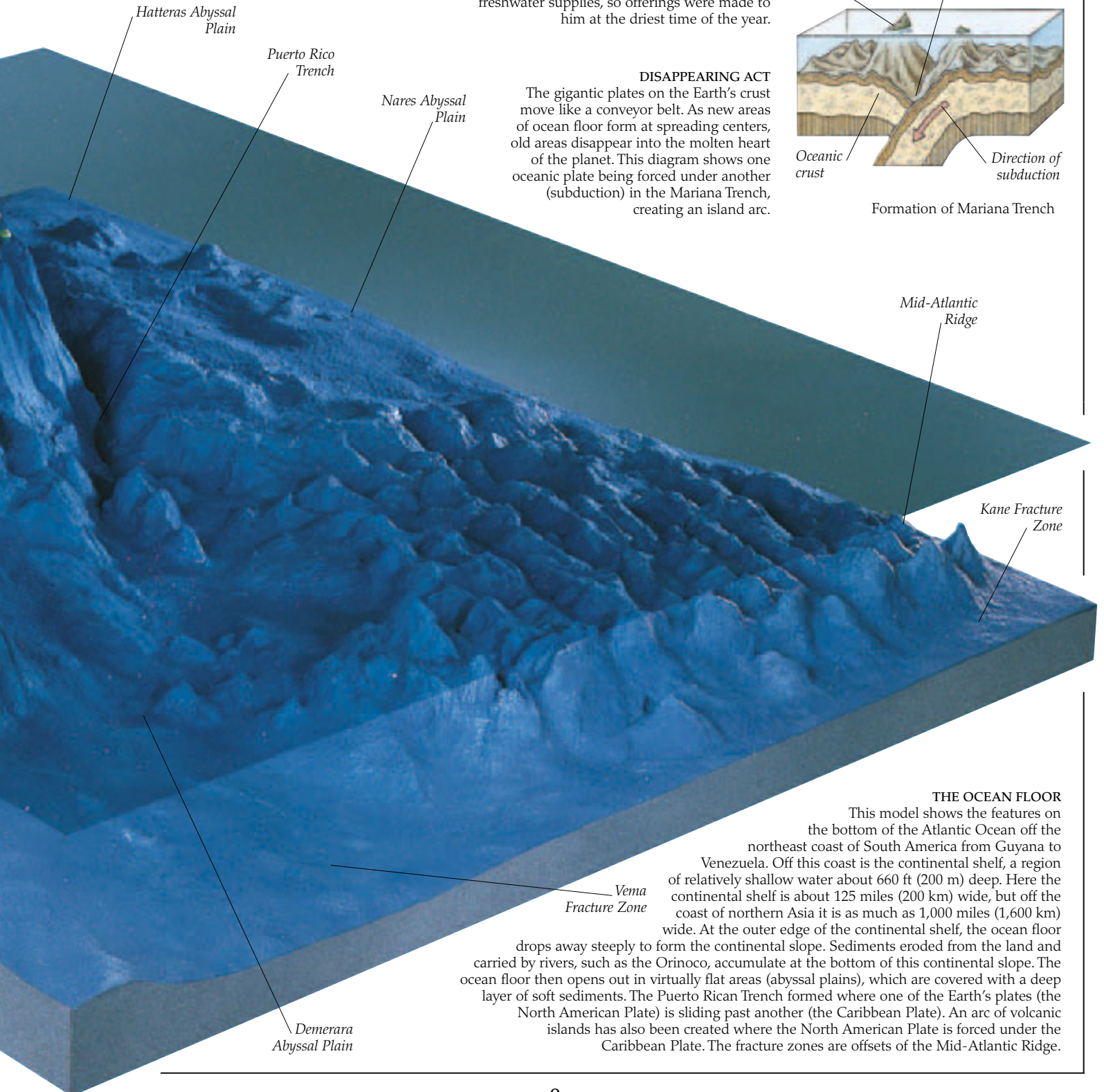
Neptune, the Roman god of the sea, is usually shown riding a dolphin and carrying a pronged spear (trident). It was thought he also controlled freshwater supplies, so offerings were made to him at the driest time of the year.

DISAPPEARING ACT

The gigantic plates on the Earth's crust move like a conveyor belt. As new areas of ocean floor form at spreading centers, old areas disappear into the molten heart of the planet. This diagram shows one oceanic plate being forced under another (subduction) in the Mariana Trench, creating an island arc.



Formation of Mariana Trench



THE OCEAN FLOOR

This model shows the features on the bottom of the Atlantic Ocean off the northeast coast of South America from Guyana to Venezuela. Off this coast is the continental shelf, a region of relatively shallow water about 660 ft (200 m) deep. Here the continental shelf is about 125 miles (200 km) wide, but off the coast of northern Asia it is as much as 1,000 miles (1,600 km) wide. At the outer edge of the continental shelf, the ocean floor drops away steeply to form the continental slope. Sediments eroded from the land and carried by rivers, such as the Orinoco, accumulate at the bottom of this continental slope. The ocean floor then opens out in virtually flat areas (abyssal plains), which are covered with a deep layer of soft sediments. The Puerto Rican Trench formed where one of the Earth's plates (the North American Plate) is sliding past another (the Caribbean Plate). An arc of volcanic islands has also been created where the North American Plate is forced under the Caribbean Plate. The fracture zones are offsets of the Mid-Atlantic Ridge.